Linking Network Structure, Modularity, and Controllability in Age-Structured Contact Networks

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Human contact networks shape how information and infectious diseases spread, yet the links between demographic mixing, mesoscale organization, and dynamical controllability remain unclear. We analyze age-stratified contact matrices from six countries across home, school, work, and other social contexts to quantify structural features (clustering, path length, efficiency), mesoscale modularity, and control-theoretic properties (average controllability, modal controllability, minimum control energy). We find that home and other settings form cohesive, navigable networks, while school networks exhibit pronounced age-assortative blocks and work networks are diffuse. Modularity is modest and consistently below strength-preserving and age-distance null expectations, indicating that apparent communities largely reflect smooth age-based mixing. Embedding each matrix as the coupling operator in a linear dynamical system reveals consistent context-dependent roles: adolescents are effective local influencers in schools (high average controllability), young adults preferentially steer slow modes in workplaces (high modal controllability), and school networks require the greatest energy to achieve global state changes. These results provide a unified, reproducible framework linking structure and modularity to controllability in demographic networks and suggest context-sensitive strategies for public-health and behavioral interventions.

social contact networks | network controllability | modularity | age-structured mixing | dynamical systems | public health

Significance Statement

Human contact networks shape how behaviors and infectious diseases spread, yet their deeper structural and dynamical properties remain underexplored. Using age-structured contact matrices from six countries across home, school, work and other social settings, we show that much of the apparent modular organization in human interactions arises from smooth age-based mixing rather than sharply defined community partitions. However, schools retain pronounced age-aligned structure, while workplaces exhibit diffuse and fragmented patterns. Extending these findings with network controllability analysis, we demonstrate that such structural and mesoscale differences strongly influence which demographic groups are most effective or costly to steer. These results establish a quantitative foundation for integrating demographic mixing, modular organization and dynamical control into epidemiological and behavioral-intervention models, informing context-sensitive public health strategies.

Human social interactions provide the channels through which information, behaviors, and infectious diseases propagate (1). Decades of work in network science have shown that such systems are not random: many real-world networks exhibit small-world structure and short characteristic path lengths, enabling rapid signal propagation across large populations (2). At the same time, heterogeneous degree distributions introduce strong asymmetries in influence and vulnerability, concentrating dynamical impact in a subset of nodes (3). Beyond these global features, clustering and modular organization further shape how robustly signals spread and how systems respond to perturbations (4).

Age-structured contact matrices provide a particularly informative representation of human interaction patterns. Widely used in infectious-disease modeling, these matrices quantify the frequency of contacts between demographic groups and thus encode the pathways along which population-level dynamics unfold (5). Recent efforts have begun to analyze these matrices explicitly as weighted networks, revealing systematic variation in clustering, path length, and mesoscale structure across countries and social settings (6). For example, comparative studies have shown that school environments are dominated by dense adolescent-to-adolescent mixing, home settings display strong age-assortative blocks spanning adjacent generations, and workplace networks are typically sparse and fragmented (7, 8). Such patterns persist across diverse geographic regions and demographic compositions, indicating that social context imposes structural constraints beyond country-specific factors (9).

These structural and community-level features are not merely descriptive. A growing literature has demonstrated

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J.Z. designed the study, performed all analyses, created all figures, and wrote the manuscript.

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that network architecture fundamentally constrains dynamical processes, including diffusion, synchronization, and epidemic spread (10, 11). Yet despite their central role in shaping population-level dynamics, far less is known about how the structural and modular properties of age-structured contact networks influence their controllability—that is, whether and how external inputs applied to specific age groups can steer system trajectories.

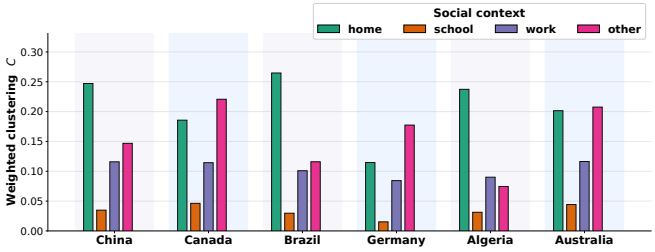
Controllability has emerged as a core concept in the study of complex networks, formalizing how network topology restricts the ability to drive a system between states (12). Subsequent work has shown that different nodes may dominate different dynamical regimes, motivating distinctions between average controllability, modal controllability, and energy-based notions of control (13, 14). These ideas have proven particularly powerful in biological and neural systems, where structural connectivity strongly shapes dynamical flexibility and control costs (15). However, their application to demographic contact networks remains limited.

In public health and behavioral intervention settings, vaccination campaigns, communication strategies, and targeted resource allocation can all be interpreted as control inputs applied to selected demographic groups. Which age groups are most effective targets, and whether certain populations are intrinsically costly to influence, likely depends on how local connectivity, global position, and modular segmentation interact within the contact network (16, 17). Yet existing studies have not systematically linked these structural features to control-theoretic measures across countries and social contexts.

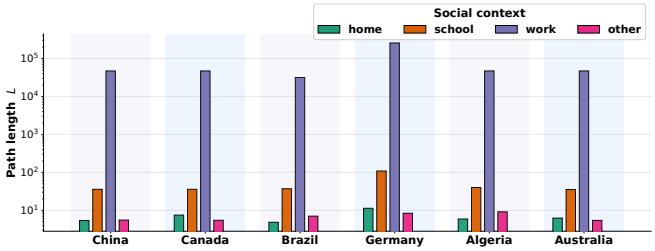
This gap motivates the central question of the present study: **How do the structural and modular properties of human contact networks shape their controllability, and which age groups are most influential or costly to control across countries and social settings?** Addressing this question requires integrating network structure, community organization, and dynamical control within a unified framework.

Hypotheses. We test four hypotheses grounded in prior work. H1: school networks exhibit higher modularity than work networks across countries. H2: adolescents display higher average controllability than other age groups in school networks. H3: young adults (ages 20–34) display higher modal controllability than other age groups in work networks. H4: school networks require higher minimum control energy than work networks.

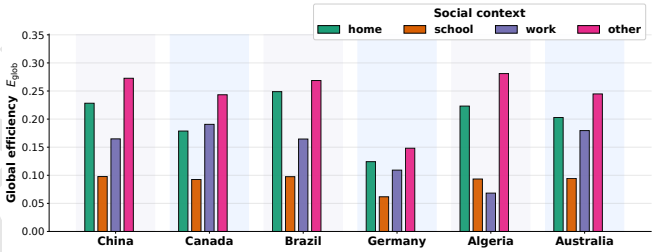
To answer it, we integrate structural analysis, modularity-based community detection, and control-theoretic modeling across age-binned contact matrices from six countries. Each matrix is treated as the coupling matrix A of a linear dynamical system $\dot{x} = Ax + Bu$, allowing observed cross-age interactions to be interpreted within a dynamical framework. We compute complementary controllability metrics, including average controllability, modal controllability, and minimum control energy, each capturing distinct dynamical roles of different age groups. By comparing these measures across home, school, work, and other social contexts and grounding their interpretation in the structural and modular patterns quantified earlier, we provide a unified assessment of how demographic mixing shapes the ease or difficulty of steering population-level dynamics.



(a) Clustering. Weighted clustering coefficient C for six countries and four social contexts. Higher C indicates denser local triangles and stronger age assortativity.



(b) Path length. Characteristic path length L computed from weighted shortest-path distances. Because values span several orders of magnitude, the vertical axis is shown on a logarithmic scale. Work networks display extremely long paths, reflecting weak cross-age ties.



(c) Global efficiency. Global efficiency E_{glob} for the same country–context combinations. Higher efficiency indicates more redundant cross-age communication pathways.

Fig. 1. Network structure across countries and social contexts. Each panel shows one structural metric: weighted clustering C (a), characteristic path length L (b), and global efficiency E_{glob} (c). Bars represent four social contexts (home, school, work, other) within each country, with alternating background shading indicating different countries. Together these measures summarize local cohesion, long-range navigability, and global communication capacity of age-structured contact networks.

Results

Network structure analysis. We quantified the structural organization of the contact networks using weighted clustering, characteristic path length, and global efficiency. The full set of values across countries and social contexts is visualized in Fig. 1.

Clustering. A consistent pattern emerges across all countries: home environments exhibit the strongest local cohesion, with clustering values typically around 0.20–0.26. Other-location networks also show moderately high clustering, while work settings display only modest levels of local structure. School networks are the most diffuse, with clustering often an order of magnitude lower than in home settings. These differences reflect the presence of dense age-adjacent interactions in households and classrooms versus more heterogeneous mixing in workplaces.

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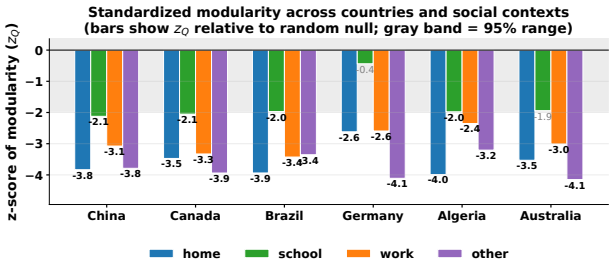

Fig. 2. Modularity and community structure across countries and contexts. Bars show Louvain modularity Q for each of the 24 age–structured contact networks. Labels indicate the number of detected communities in the consensus partition. Home and school networks exhibit the strongest modular organization, while work and other settings display weaker and more diffuse mesoscale structure.

Path length. The contrast in characteristic path length is even more pronounced. Home, school, and other networks exhibit relatively short weighted paths, usually between five and ten steps for home and other settings, and in the mid-thirties for school environments. In stark contrast, all six countries show extremely long paths in the work layer. In several cases, the weighted path length exceeds 10^4, and for Germany it reaches well above 10^5. This inflation arises from very small cross-age contact weights, which dramatically increase the weighted distances and highlight the strong age stratification of workplace interactions. The all-location aggregated matrices, by contrast, consistently produce the shortest paths due to the superposition of multiple interaction layers. Across countries, work networks have longer paths than school networks in 6/6 cases (paired sign test, two-sided $p = 0.031$).

Global efficiency. Global efficiency follows the inverse of these patterns. Networks in home and other settings support relatively efficient communication across age groups. School networks are less efficient, and work networks are the least efficient in every country, sometimes by factors of two to three relative to home settings. The aggregated all-location networks show the highest efficiency overall, consistent with their short paths and composite structure. Across countries, home networks have higher global efficiency than work networks in 6/6 cases (paired sign test, two-sided $p = 0.031$).

Taken together, these results show that home and other settings form cohesive and relatively navigable networks, school contexts are more diffuse but still globally reachable, and work settings are highly fragmented with limited cross-age connectivity. These structural patterns shape how easily control inputs can propagate through the system, motivating the modularity and controllability analyses that follow.

Modularity analysis. We next examine whether age–structured contact networks exhibit mesoscale organization that could influence how control signals propagate through the linear dynamics $\dot{x} = Ax + Bu$. Modular communities are expected to constrain controllability by trapping input energy within densely connected age groups, increasing the energy required to steer the system across modules.

Stable but weak modular structure. For each country and social context, we ran $K = 150$ Louvain initializations and obtained the distribution of modularity values $Q^{(r)}$ (Fig. 3). Across all 24 networks, the distributions are narrow, indicating that the community structure is statistically reproducible despite the stochasticity of Louvain. However, the absolute modularity levels are modest: home and school networks reach $Q \approx 0.10$–0.22, whereas work networks show $Q \approx 0$, reflecting an absence of clear modular segregation. This already suggests that control energy will not be strongly confined by community boundaries in work settings but may be more localized in home and school environments.

Comparison to null models with strength preservation. To evaluate whether the observed modular organization exceeds what is expected from degree and strength sequences alone, we computed standardized modularity scores z_Q relative to weighted Chung–Lu null networks (Fig. 2). All networks exhibit *negative* z_Q (-4.1 to -0.4), implying that their community structure is *weaker* than what would be expected from random networks with the same strength distribution. This is consistent with age–structured contact matrices being highly diffuse, with many cross-age interactions that reduce mesoscale separability. From a controllability perspective, this predicts more homogeneous influence propagation and fewer strongly isolated subpopulations.

Spatial age-distance nulls. This result is critical because it separates two mechanisms that can both produce diagonal block structure in contact matrices: smooth decay with age distance versus genuinely discrete community segregation. The spatial null therefore provides the most interpretable baseline for assessing whether Louvain communities reflect institution-driven mixing beyond demographic proximity. Modularity also falls below expectations under spatial age-distance nulls fitted to exponential decay. Across all country–context networks, consensus modularity is lower than the spatial null mean (24/24 instances; paired sign test, two-sided $p < 10^{-6}$), indicating that much of the apparent community structure can be explained by smooth age-based mixing.

Consensus communities reinforce context-specific patterns. We aggregated the K partitions into a co-association matrix, thresholded at $\tau = 0.6$, and extracted consensus communities $\hat{g}$. Reordered matrices with community boundaries are shown in Fig. 4. A consistent diagonal block appears in every country, confirming universal age-assortative mixing. However , the *sharpness* of this block varies strongly by context: home and school networks produce 2–3 well-defined communities, while work and "other" networks fragment into 4–6 diffuse modules with weak boundaries. This matches the modularity scores and implies that controllability will vary by setting: tighter modules in home/school may require more energy to move the system between age groups, whereas work networks may allow broad, low-cost influence spread.

In general, age–structured contact networks contain a reproducible but relatively weak modular organization, strongly shaped by the social context. The consensus communities extracted here serve as the intermediate layer linking structural connectivity to the controllability outcomes analyzed next (average controllability, modal controllability, and minimum control energy).

Interpretation of community composition. Beyond their strength, the detected communities exhibit distinct and interpretable age compositions that vary systematically by social context. In school networks, consensus partitions typically separate children (0–9), adolescents (10–19), and young adults into two or three contiguous age blocks, corresponding closely to primary,

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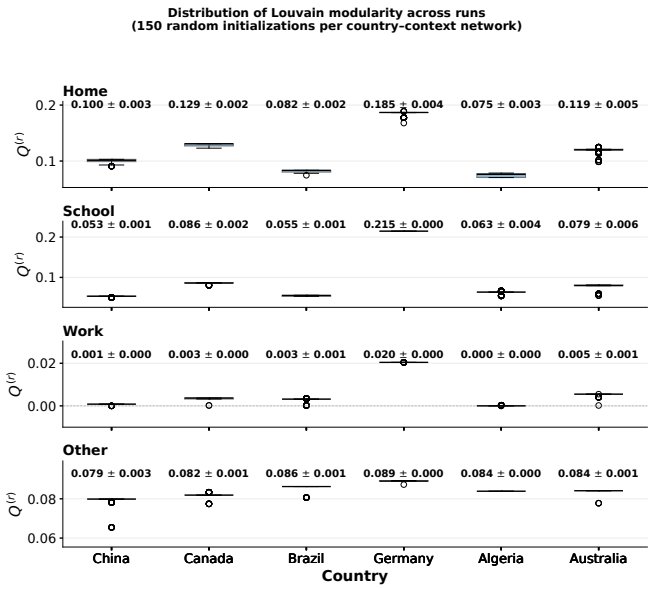

Fig. 3. Distribution of Louvain modularity across runs. Boxplots show the distribution of modularity values $Q^{(r)}$ from $K = 150$ Louvain initializations for each country and social context. Home and school networks exhibit moderate, stable modularity, whereas work networks have $Q^{(r)}$ values close to zero.

secondary, and post-secondary or staff-associated interaction layers. These age-aligned communities are sharp and stable across countries, indicating that institutional scheduling and classroom structure produce strongly segregated mesoscale organization.

Home networks display coarser but socially meaningful communities, often grouping children with middle-aged adults and older adults into overlapping modules that reflect household and intergenerational mixing. These communities are less sharply bounded than in schools but remain consistent across countries, suggesting shared family-based contact patterns.

In contrast, work networks fragment into multiple diffuse communities spanning broad working-age ranges (approximately 25–64), with weak and irregular boundaries. Rather than discrete age blocks, these modules reflect heterogeneous occupational mixing and cross-generational interaction, resulting in mesoscale structure that is reproducible but poorly segregated. This qualitative difference in community composition explains why work networks exhibit low modularity yet remain highly controllable in later analyses.

Controllability analysis. To examine how structural and modular properties shape the dynamical influence of different age groups, we evaluate three control-theoretic metrics derived from the linear system $\dot{x} = Ax + Bu$. All analyzes use the loop-free symmetrized matrix A described above so that controllability reflects meaningful cross-age coupling rather than survey-level noise.

Average controllability. Figure 5 presents country-resolved heat maps of average controllability by node under full control input $B = I_{16}$. In all six countries, two consistent patterns appear.

First, *school* networks universally show prominent peaks among adolescents (ages 10–19), reflecting the dense, age-homogeneous mixing that amplifies their ability to distribute small perturbations across the system. This adolescent dominance appears in China, Canada, Brazil, Algeria, and Australia, and is especially pronounced in Germany where the 15–19 bin is the strongest AC node across all country–context pairs.

Second, *work* networks consistently shift their AC maxima to middle-aged adults (ages 35–49). This pattern is highly visible in Canada, Brazil, Germany, and Australia, suggesting that mid-career adults serve as structural bridges across working-age strata. Home and other contexts display weaker gradients, but still exhibit a mild rise in AC among young or middle-aged adults, reflecting household and community interaction structures.

Modal controllability. Figure 6 reports modal controllability, which captures each node's ability to influence difficult-to-reach dynamical modes associated with weakly coupled eigenvectors. Unlike average controllability, a distinct cross-country regularity emerges: *young adults (20–34)* dominate modal controllability in nearly every country's work network. This pattern is strikingly stable across China, Canada, Brazil, Germany, and Australia, indicating that young adults are the key drivers for steering slow or weak modes of the system in economically structured environments.

In contrast, school networks in all countries show uniformly suppressed modal controllability—adolescents who dominate average controllability do *not* effectively control slow modes. This reflects the strong modular blocks detected earlier: tightly knit school communities amplify local influence but restrict global, mode-spanning control. Other contexts show more heterogeneous patterns but still place their modal peaks mostly in early to mid adulthood.

Minimum control energy. Figure 7 shows the minimum energy required to actuate the system under full control. A robust cross-country pattern emerges: *school networks require the highest energy*, while *work networks consistently require the lowest*. This ordering appears in all six countries and mirrors the modularity results from Section 4.2. Highly assortative, age-segregated school networks trap energy within local blocks, making global steering expensive. Work networks, with weaker modularity and more cross-age ties, permit long-range propagation at lower energy cost. Home and other settings fall in between, with modest but consistent country-level variability. Across countries, school networks require higher energy than work networks in 6/6 cases (paired sign test, two-sided $p = 0.031$).

Synthesis and implications. Taken together, these analyzes reveal a coherent, cross-country pattern linking network structure, modular organization, and dynamical influence. (1) Adolescents universally dominate *average controllability* in schools, reflecting dense, homogeneous mixing that enhances local responsiveness. (2) Young adults dominate *modal controllability* in work networks across all six countries, revealing their central role in steering slow or weak system modes. (3) School networks require the highest *minimum control energy*, while work networks are the easiest to steer—directly reflecting the contrast between strong school modularity and weaker work modularity.

These findings provide a clear answer to our guiding question: **the structural and modular properties of human contact networks fundamentally shape their controllability, and the most influential or costly-to-control age groups shift systematically with social**

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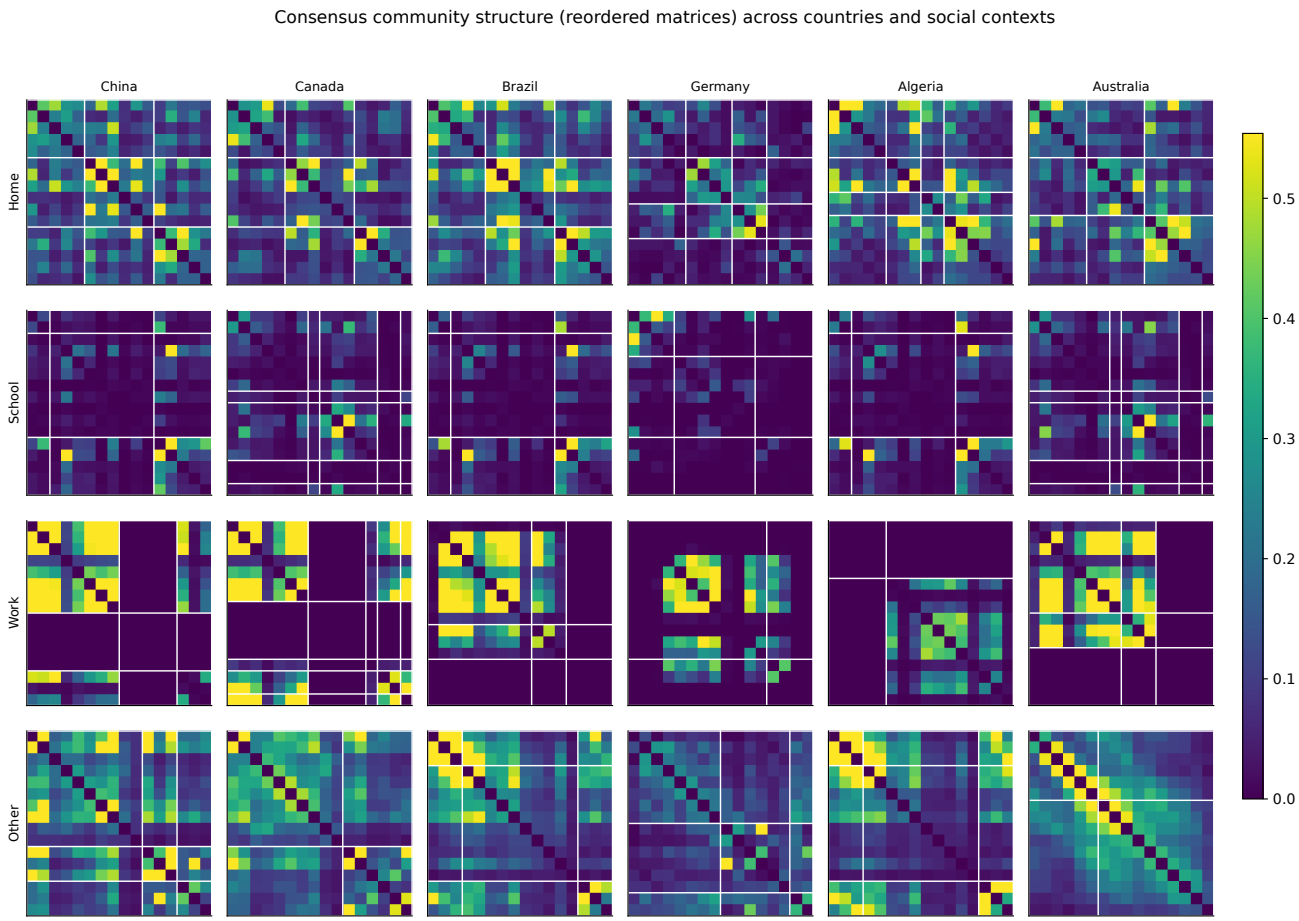

Fig. 4. Consensus community structure across countries and social contexts. Each panel shows a 16×16 age-binned contact matrix reordered by the consensus community labels $\hat{g}$ for a given country (columns) and social context (rows). Color indicates contact intensity, and white lines mark community boundaries. Home and school networks exhibit strong, age-assortative blocks, whereas work and other networks show weaker, more fragmented community structure.

context. Age-aligned modularity amplifies adolescent influence in school settings but restricts their global control power, while the broader, cross-age mixing of work networks enables young and middle-aged adults to steer the system efficiently with substantially lower energy cost.

Discussion

This study integrates structural analysis, consensus community detection, and network controllability to reveal how age–structured contact networks differ across countries and social settings, and how these differences shape the dynamical influence of specific demographic groups. Rather than being governed by a single structural property, controllability emerges from the joint interaction of clustering, age-assortative modularity, and the density of cross-age connections. This systems perspective aligns with foundational work showing that network topology constrains the capacity to steer complex systems (17, 18).

Across all six countries, we observe a coherent structural hierarchy: home and other contexts form moderately clustered and relatively efficient networks, school networks exhibit the strongest age-aligned modular blocks, and work networks display diffuse, weakly assortative patterns. These findings extend earlier analyses of social and epidemiological networks (7, 19) by showing that mesoscale structure is not solely a by-product of smooth age-mixing decay but is shaped by institutional and behavioral constraints characteristic of each environment.

Embedding these structural differences in a linear control framework provides a dynamical interpretation. Average controllability highlights adolescents in schools and middle-aged adults in workplaces as rapid-influence conduits, consistent with the idea that locally well-connected nodes facilitate low-energy, short-horizon transitions (13, 20). Modal controllability identifies young adults, particularly in work settings, as key access points for slow or weakly coupled modes, a pattern consistent with the role of eigenstructure in shaping high-order dynamics (10, 16). These complementary patterns illustrate how distinct age groups dominate different dynamical regimes.

Minimum control energy further clarifies the impact of modular segmentation. School networks require substantially greater energy to steer than work networks, reflecting the fact that strong, cohesive blocks trap influence within modules and hinder long-range transitions. This finding parallels theoretical results linking modularity, coupling strength, and the energetic cost of state transitions (15, 21). By contrast, work networks, with weaker community structure, permit more

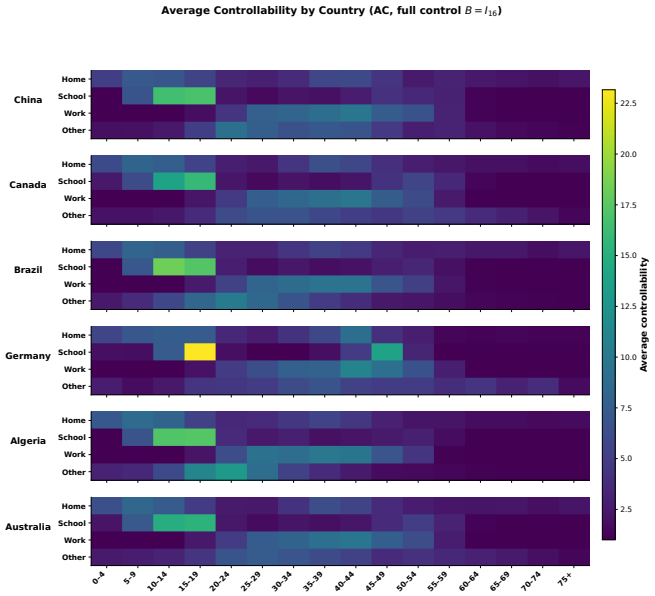

Fig. 5. Average controllability (AC) across countries, age groups, and social contexts.

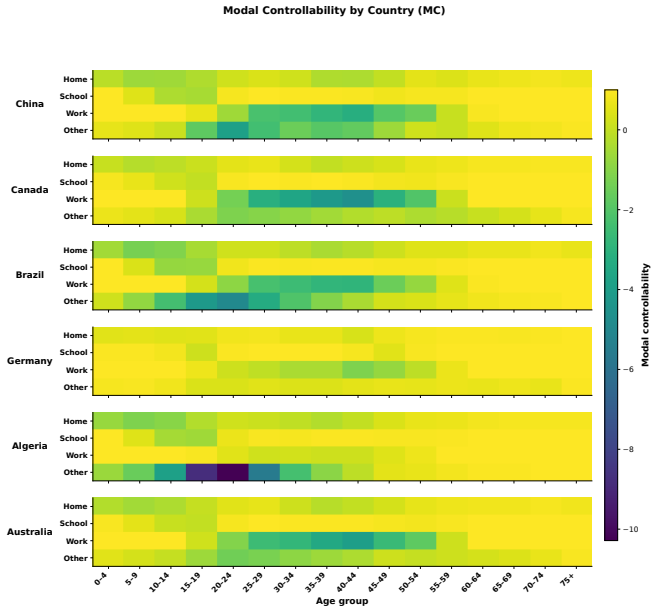

Fig. 6. Modal controllability (MC) across countries, age groups, and social contexts.

efficient single-entry interventions that spread broadly through diffuse connection pathways.

Taken together, these results establish a cross-country robust principle: the controllability of demographic contact networks depends jointly on local density, mesoscale modularity, and cross-age coupling. Adolescents, young adults, and middle-aged groups occupy distinct dynamical roles depending on the social setting, shaping which age groups are most influential or costly to control. This insight provides a quantitative foundation for targeted public-health interventions, optimized resource allocation, and dynamical models of influence or behavior spread (22, 23).

Several limitations should be noted. Diary-based contact matrices may underreport short or incidental interactions, age binning aggregates heterogeneous behaviors, and the linear, time-invariant model captures only local approximations to real social dynamics (24). Louvain community detection is heuristic and may miss alternative partitions, though consensus aggregation improves reliability. Finally, symmetrization and loop-removal simplify directed or repeated interactions, which may influence controllability in real populations.

An additional limitation is that our analysis treats home, school, work, and other contexts as distinct networks rather than as a single, aggregated contact structure. In reality, population-level influence and intervention strategies often operate across multiple social layers simultaneously, and the superposition of these contexts may yield qualitatively different controllability properties. For example, aggregating layers could weaken age-aligned modular boundaries observed in schools while introducing new cross-context pathways mediated by households or workplaces. Exploring how context-specific controllability patterns combine into a society-wide contact network represents an important direction for future work, particularly for modeling interventions that act across multiple social domains rather than within a single setting.

Despite these limitations, the convergence of structural, modular, and dynamical evidence reveals a consistent organizing principle: the ease of controlling population-level dynamics is fundamentally shaped by the interplay between assortative mixing, modular segmentation, and the availability of cross-age interactions. This framework enables richer models of demographic intervention strategies and highlights how age structure shapes not only epidemic risk but also the dynamical landscape of influence in human societies.

Materials

Contact matrices are publicly available from Prem et al. (2017) (6). Analysis scripts and figure-generation code are available at https://github.com/Junfei-Z/age-structured-contact-controllability. All parameters, seeds, and environment details required to reproduce Tables and Figures are documented in the repository README.

Methods

An overview of the full analysis pipeline, including contact-matrix inputs, network structure quantification, modularity analysis, and controllability evaluation, is schematically illustrated in Fig. 8. Each component of the pipeline is described in detail in the following subsections.

Data preprocessing and graph construction. We analyze age–structured social contact matrices (16×16) from six countries (China, Canada, Brazil, Germany, Algeria, and Australia) across four social contexts (home, school, work, other) (6). The raw matrices M are derived from self-reported surveys and often exhibit asymmetric sampling noise ($M_{ij} \neq M_{ji}$), even though real-world face-to-face contact is inherently reciprocal. To obtain a realistic and analytically consistent representation of inter-age interactions, we construct a symmetrized, loop–free weight matrix

$$W = \tfrac{1}{2}\left(M + M^{\top}\right), \qquad W_{ii} = 0, \qquad [1]$$

and standardize entries to contacts per person per day.

Constructing a symmetrized, loop-free weight matrix is essential for both network and control-theoretic analysis.

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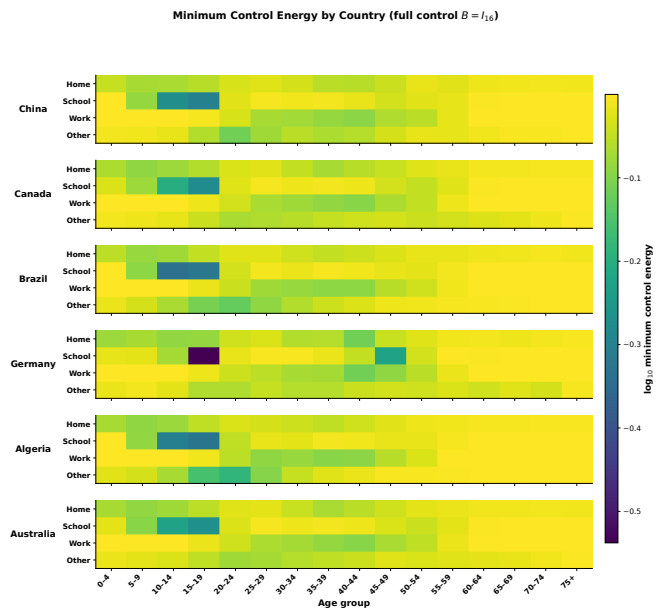

Fig. 7. Minimum control energy $E_{\min}$ (full control $B = I_{16}$), across countries, age groups, and social contexts.

Symmetrization removes survey-induced asymmetry ($M_{ij} \neq M_{ji}$) and provides a realistic representation of reciprocal contact. Zeroing the diagonal eliminates within-age-bin mixing, which does not constitute graph edges and would otherwise distort measures such as total edge weight, clustering, and modularity. Self-loops are also handled inconsistently across Louvain implementations, yielding algorithm-dependent communities. From a dynamical-systems perspective, diagonal entries of the coupling matrix encode intrinsic decay rather than self-contact; retaining empirical diagonal values would alter the eigenvalue spectrum, affect stability, and modify the controllability Gramian W_c. A symmetric, zero-diagonal adjacency ensures that all structural and controllability metrics reflect cross-age coupling only.

The processed matrix W defines a weighted, undirected contact network $G = (V, E)$ with $|V| = 16$ nodes representing age groups. For all structural, modular, and control analyses, we take

$$A = W, \qquad [2]$$

so that A simultaneously serves as the adjacency matrix for network characterization and as the coupling matrix for the linear dynamical model used in controllability analysis. Node strength is computed as $k_i = \sum_j A_{ij}$ and total edge weight as $2m = \sum_{ij} A_{ij}$. This unified construction provides a consistent foundation for quantifying how structural and modular organization shape the controllability of age-structured contact patterns across countries and social contexts.

Network structure analysis. To quantify how age-structured contact patterns influence the controllability of the system, we compute three weighted network measures from the adjacency matrix A: clustering, characteristic path length, and global efficiency. These features capture local connectivity patterns and global communication capacity, both of which shape how control inputs propagate through the linear model $\dot{x} = Ax + Bu$.

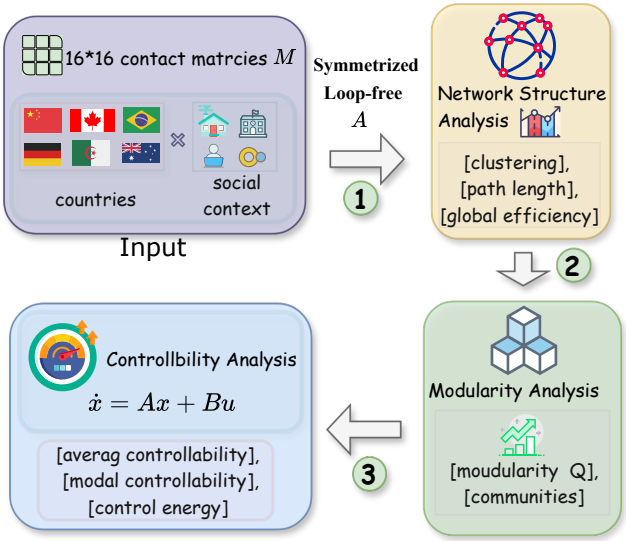

Fig. 8. Workflow for contact-network structure and controllability analysis. Age-structured contact matrices (16×16) from six countries and four social contexts are converted into weighted networks to compute basic structural measures (degree, clustering, path length) and Louvain modularity. These network representations then serve as the coupling matrix A in a linear control model $\dot{x} = Ax + Bu$, from which we derive average and modal controllability, and minimum control energy to compare control difficulty and critical age groups across settings.

Weighted clustering coefficient (C). Following Barrat et al. (25), the local clustering of node i is

$$C_i = \frac{1}{s_i(k_i - 1)} \sum_{j,h} \frac{w_{ij} + w_{ih}}{2} a_{ij} a_{ih} a_{jh}, \qquad [3]$$

where k_i is degree, s_i is strength, and a_{ij} indicates edge presence. Higher clustering reflects tighter local triads among nearby age groups; such cohesion tends to confine interactions locally and reduces the ability of control inputs to spread broadly.

Characteristic path length (L). Weighted shortest-path distances d^*_{ij} are computed using reciprocal edge weights. The characteristic path length is

$$L = \frac{1}{N(N-1)} \sum_{i \neq j} d^*_{ij}, \qquad [4]$$

where larger L reflects fragmented, age-segregated mixing that lowers propagation efficiency and increases the control energy required to steer the system.

Global efficiency (E_{glob}). Efficiency captures the inverse of weighted shortest-path distances (26):

$$E_{\text{glob}} = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{d^*_{ij}}. \qquad [5]$$

High global efficiency indicates robust cross-age communication and redundant pathways, predicting higher average controllability and lower minimum control energy.

Together, these structural metrics characterize the overall organization of each country–context network. Because control inputs propagate along the edges of A, differences in clustering, path length, and efficiency determine the ease with which external signals can spread across age groups. These structural insights provide the basis for the modularity and controllability analyses presented in the following subsections.

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Modularity analysis. To quantify mesoscale organization in age-structured contact networks, we identify communities by maximizing the weighted Newman–Girvan modularity with resolution parameter γ:

$$Q(\gamma) \;=\; \frac{1}{2m}\sum_{i,j}\left(A_{ij} - \gamma\,\frac{k_i k_j}{2m}\right)\,\delta(g_i, g_j), \qquad [6]$$

where g_i denotes the community label of node i and k_i is the node strength(27). Because modularity reflects the density of within-community connections relative to null expectations, it provides a mesoscale structural lens that is directly relevant for controllability: tightly clustered modules can trap control energy, while inter-module connectors facilitate global influence.

Louvain community detection. We apply the Louvain (GenLouvain) heuristic with $K = 150$ random initializations and $\gamma = 1$ (28). Each run yields a partition $g^{(r)}$ and modularity value $Q^{(r)}$, enabling assessment of partition stability.

Consensus partition. We summarize community structure via the 16×16 co-association matrix

$$C_{ij} = \frac{1}{K}\sum_{r=1}^{K}\mathbf{1}\{g_i^{(r)} = g_j^{(r)}\}, \qquad [7]$$

thresholded at $\tau = 0.6$ and reclustered with Louvain to obtain a consensus partition $\hat{g}$ and modularity $\hat{Q}$ (29). Partition uncertainty is quantified by the mean pairwise Variation of Information across runs.

Null models. We evaluate whether the observed modular structure exceeds two increasingly restrictive baselines.

Strength-preserving weighted Chung–Lu null. We first generate $R = 500$ random weighted networks that preserve the empirical node-strength sequence in expectation. Specifically, the expected weight between nodes i and j is

$$P_{ij} = \frac{k_i k_j}{2m}, \qquad [8]$$

where $k_i = \sum_j A_{ij}$ and $2m = \sum_{ij} A_{ij}$ (30). For each null realization, we run Louvain to obtain $Q_{\text{null}}^{(r)}$ and compute the standardized score $z_Q = (\hat{Q} - \mu_{\text{null}})/\sigma_{\text{null}}$.

Spatial age-distance null. Because age-structured contact intensities typically decay smoothly with age separation, modularity can appear even in the absence of true community segregation. To control for this effect, we construct a spatial null in which expected weights depend only on the age distance between bins:

$$\mathbb{E}[A_{ij}] = \theta_0 e^{-\alpha\Delta_{ij}}, \qquad \Delta_{ij} = |a_i - a_j|, \qquad [9]$$

where a_i is the midpoint of age bin i, and θ_0, α are fitted by least squares on the empirical matrix. To ensure a fair comparison, we rescale the resulting expected-weight matrix using iterative proportional fitting so that the null matches the empirical strength sequence while retaining the exponential distance decay. We then generate $R = 200$ spatial-null networks by sampling weights around these expectations and compute a spatial significance score $z_Q^{\text{spa}} = (\hat{Q} - \mu_{\text{spa}})/\sigma_{\text{spa}}$. This spatial baseline directly tests whether detected communities reflect mesoscale segregation beyond what is explained by smooth age proximity alone.

Together, these modularity analyses reveal whether age-structured contact networks exhibit mesoscale segregation beyond what is expected from degree distributions or age proximity alone. Because modular organization constrains how control energy propagates across modules in the dynamical system $\dot{x} = Ax + Bu$, the resulting community structure serves as a key intermediary between local structural characteristics and controllability outcomes examined in the following subsections.

Controllability analysis. To quantify how structural and modular organization shape the ease of influencing age-structured contact dynamics, we model the system using a standard linear time-invariant network control framework. Each age group is represented by a state variable $x_i(t)$, and the evolution of the system is governed by

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad [10]$$

where A is the loop-free symmetrized contact matrix derived in the preprocessing step, $u(t)$ is a vector of external control inputs, and B specifies the placement of control nodes. Because A directly encodes cross-age coupling strength, this model captures how interventions applied to specific age groups propagate throughout the contact network.

We evaluate controllability under two conditions: (i) full control, $B = I_{16}$, where each age group receives an independent input; and (ii) single-node control, $B = e_i$, where a unit input is applied to age group i only. The latter allows identification of age groups that serve as effective controllers in each country–context network.

Average controllability (AC). Average controllability quantifies how easily small, distributed perturbations to a node can move the system into nearby states. We compute AC from the Gramian

$$W_c = \int_0^\infty e^{At}BB^\top e^{A^\top t}\,dt, \qquad [11]$$

and interpret higher AC as more efficient spread of control energy.

Modal controllability (MC). Modal controllability characterizes a node's ability to influence difficult-to-reach eigenmodes of A. Nodes with high MC steer slow or weakly coupled modes and often correspond to peripheral or age-isolated groups.

Minimum control energy. We quantify the cost of steering the system by the minimum energy to reach a target state x_T over horizon T:

$$E_{\min} = x_T^\top W_c^{-1} x_T, \qquad [12]$$

with lower values indicating networks and age groups that are easier to influence.

Together, these controllability metrics connect the structural features (clustering, path length, and efficiency) and mesoscale organization (community structure) of contact networks to their dynamical ease or difficulty of control. This framework enables systematic identification of age groups that act as influential controllers and those that are costly to steer, and reveals how these patterns vary across countries and social contexts. In doing so, it directly addresses our central question of how structural and modular properties jointly shape the controllability of human contact networks.

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Table 1. Supplementary Table 1. Summary of modularity results by country and social context. Values are consensus modularity $\hat{Q}$, stability (VI; mean variation of information across Louvain runs), and the mean/SD of modularity under the strength-preserving random null (μ_{null}, σ_{null}) and the age–distance spatial null (μ_{spa}, σ_{spa}). z-scores used in the main text are computed as $z_Q = (\hat{Q} - \mu_{\text{null}})/\sigma_{\text{null}}$ and $z_Q^{\text{spa}} = (\hat{Q} - \mu_{\text{spa}})/\sigma_{\text{spa}}$.

Country	Context	$\hat{Q}$	VI	μ_{null}	σ_{null}	μ_{spa}	σ_{spa}
China	home	0.103	0.501	0.391	0.075	0.237	0.038
China	school	0.054	0.367	0.397	0.162	0.548	0.094
China	work	0.001	0.081	0.247	0.080	0.475	0.053
China	other	0.080	0.060	0.326	0.065	0.215	0.037
Canada	home	0.131	0.469	0.431	0.086	0.281	0.050
Canada	school	0.087	0.342	0.422	0.163	0.555	0.086
Canada	work	0.004	0.472	0.229	0.068	0.471	0.054
Canada	other	0.083	0.202	0.355	0.069	0.231	0.037
Brazil	home	0.083	0.799	0.366	0.072	0.221	0.039
Brazil	school	0.056	0.542	0.378	0.164	0.539	0.089
Brazil	work	0.003	0.263	0.252	0.072	0.481	0.049
Brazil	other	0.086	0.000	0.314	0.068	0.214	0.039
Germany	home	0.189	0.224	0.490	0.115	0.373	0.076
Germany	school	0.215	0.000	0.311	0.220	0.372	0.149
Germany	work	0.020	0.096	0.289	0.104	0.453	0.062
Germany	other	0.089	0.356	0.466	0.092	0.315	0.062
Algeria	home	0.079	0.802	0.384	0.077	0.236	0.044
Algeria	school	0.067	0.367	0.394	0.166	0.530	0.094
Algeria	work	0.000	0.264	0.330	0.140	0.461	0.082
Algeria	other	0.084	0.000	0.286	0.063	0.192	0.038
Australia	home	0.125	0.677	0.407	0.080	0.264	0.050
Australia	school	0.081	0.408	0.418	0.173	0.557	0.089
Australia	work	0.005	0.183	0.239	0.078	0.478	0.054
Australia	other	0.084	0.000	0.355	0.065	0.232	0.040

Notes: VI values close to 0 indicate highly stable consensus partitions across Louvain initializations. Small negative scientific-notation values in the source CSV (e.g., -6.9×10^{-12}) are shown as 0.000 due to rounding.

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1. R Pastor-Satorras, A Vespignani, Epidemic spreading in scale-free networks. *Phys. review letters* **86**, 3200 (2001).
2. DJ Watts, SH Strogatz, Collective dynamics of 'small-world' networks. *Nature* **393**, 440–442 (1998).
3. AL Barabási, R Albert, Emergence of scaling in random networks. *Science* **286**, 509–512 (1999).
4. DS Bassett, O Sporns, Network neuroscience. *Nat. neuroscience* **20**, 353–364 (2017).
5. J Mossong, et al., Social contacts and mixing patterns relevant to the spread of infectious diseases. *PLoS medicine* **5**, e74 (2008).
6. K Prem, AR Cook, M Jit, Projecting social contact matrices in 152 countries using contact surveys and demographic data. *PLOS Comput. Biol.* **13**, e1005697 (2017).
7. P Expert, TS Evans, VD Blondel, R Lambiotte, Uncovering space-independent communities in spatial networks. *Proc. Natl. Acad. Sci.* **108**, 7663–7668 (2011).
8. O Boldea, A Alipoor, S Pei, J Shaman, G Rozhnova, Age-specific transmission dynamics of sars-cov-2 during the first 2 years of the pandemic. *PNAS nexus* **3**, pgae024 (2024).
9. V Engeli, S Roussos, N Demiris, A Hatzakis, V Sypsa, Social contact patterns and age mixing before and during covid-19 pandemic, greece, january 2020–october 2021. *Emerg. infectious diseases* **31**, 75 (2025).
10. R Pastor-Satorras, A Vázquez, A Vespignani, Dynamical and correlation properties of the internet. *Phys. review letters* **87**, 258701 (2001).
11. ME Newman, Spread of epidemic disease on networks. *Phys. review E* **66**, 016128 (2002).
12. YY Liu, JJ Slotine, AL Barabási, Controllability of complex networks. *nature* **473**, 167–173 (2011).
13. S Gu, et al., Controllability of structural brain networks. *Nat. communications* **6**, 8414 (2015).
14. G Lindmark, C Altafini, Minimum energy control for complex networks. *Sci. reports* **8**, 3188 (2018).
15. S Gu, et al., The energy landscape of neurophysiological activity implicit in brain network structure. *Sci. reports* **8**, 2507 (2018).
16. N Monshizadeh, K Camlibel, H Trentelman, Strong targeted controllability of dynamical networks in *2015 54th IEEE Conference on Decision and Control (CDC)*. (IEEE), pp. 4782–4787 (2015).
17. F Liu, AS Morse, Structural controllability of linear systems in *2017 IEEE 56th Annual Conference on Decision and Control (CDC)*. (IEEE), pp. 3588–3593 (2017).
18. F Pasqualetti, S Zampieri, F Bullo, Controllability metrics and algorithms for complex networks. *arXiv preprint arXiv:1308.1201* (2013).
19. V Gemmetto, A Barrat, C Cattuto, Mitigation of infectious disease at school: targeted class closure vs school closure. *BMC infectious diseases* **14**, 695 (2014).
20. M Rubinov, O Sporns, Weight-conserving characterization of complex functional brain networks. *Neuroimage* **56**, 2068–2079 (2011).
21. S Gu, et al., Optimal trajectories of brain state transitions. *NeuroImage* **148**, 305–317 (2017).
22. R Pastor-Satorras, C Castellano, P Van Mieghem, A Vespignani, Epidemic processes in complex networks. *Rev. modern physics* **87**, 925–979 (2015).
23. TC Schelling, Dynamic models of segregation. *J. mathematical sociology* **1**, 143–186 (1971).
24. HJ Ahn, B Hassibi, Global dynamics of epidemic spread over complex networks in *52nd IEEE Conference on Decision and Control*. (IEEE), pp. 4579–4585 (2013).
25. A Barrat, M Barthélemy, R Pastor-Satorras, A Vespignani, The architecture of complex weighted networks. *Proc. Natl. Acad. Sci.* **101**, 3747–3752 (2004).
26. V Latora, M Marchiori, Efficient behavior of small-world networks. *Phys. Rev. Lett.* **87**, 198701 (2001).
27. ME Newman, M Girvan, Finding and evaluating community structure in networks. *Phys. review E* **69**, 026113 (2004).
28. VD Blondel, JL Guillaume, R Lambiotte, E Lefebvre, Fast unfolding of communities in large networks. *J. statistical mechanics: theory experiment* **2008**, P10008 (2008).
29. A Lancichinetti, S Fortunato, Consensus clustering in complex networks. *Sci. reports* **2**, 336 (2012).
30. F Chung, L Lu, The average distances in random graphs with given expected degrees. *Proc. Natl. Acad. Sci.* **99**, 15879–15882 (2002).

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